PRECISION LATHE X2

OPERATIONS & MAINTENANCE

PART NO: LATHE-002-SYS-X

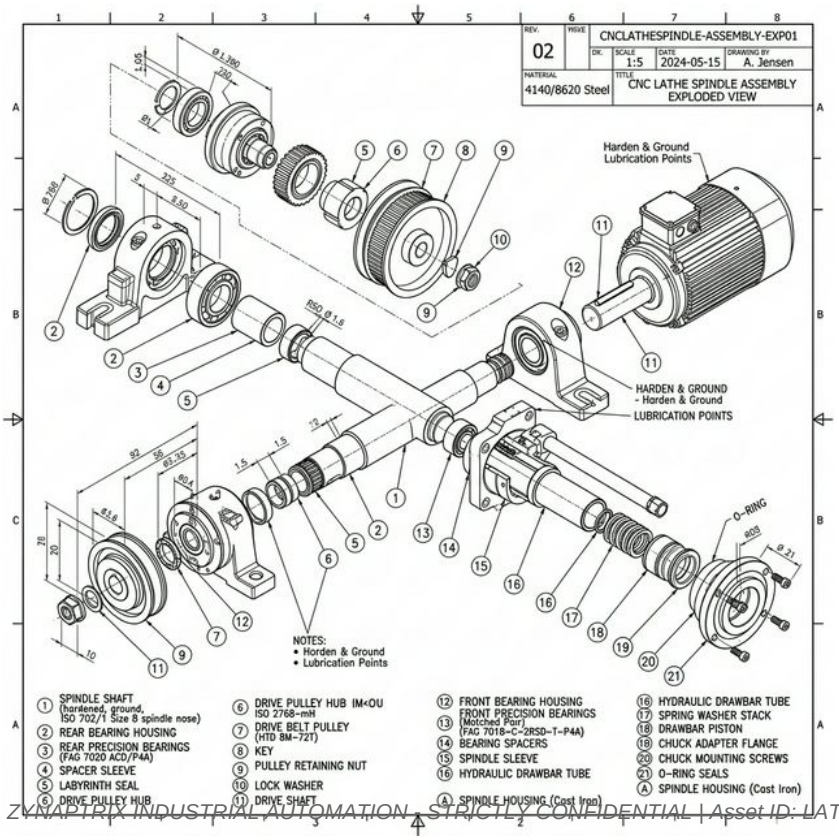

FIG 1: COMPLETE SYSTEM ASSEMBLY

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1.0 Safety & Compliance (Section 1.1)

[!] DANGER

HIGH VOLTAGE: 480VAC PRESENT. Lockout/Tagout (LOTO) procedure mandatory before removing service interlocks. Failure to comply will result in system purge.

Zynaptrix equipment complies with ISO-12100 safety of machinery. The operator must verify that all shielding is engaged before activating the primary drive sequence.

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1.0 Safety & Compliance (Section 1.2)

[!] WARNING

THERMAL HAZARD: Core temperatures may exceed 600C during shutdown phase. Allow 45 minutes for internal cooling cycle.

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1.0 Safety & Compliance (Section 1.3)

[!] DANGER

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1.0 Safety & Compliance (Section 1.4)

[!] WARNING

THERMAL HAZARD: Core temperatures may exceed 600C during shutdown phase. Allow 45 minutes for internal cooling cycle.

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1.0 Safety & Compliance (Section 1.5)

[!] DANGER

HIGH VOLTAGE: 480VAC PRESENT. Lockout/Tagout (LOTO) procedure mandatory before removing service interlocks. Failure to comply will result in system purge.

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1.0 Safety & Compliance (Section 1.6)

[!] WARNING

THERMAL HAZARD: Core temperatures may exceed 600C during shutdown phase. Allow 45 minutes for internal cooling cycle.

Zynaptrix equipment complies with ISO-12100 safety of machinery. The operator must verify that all shielding is engaged before activating the primary drive sequence.

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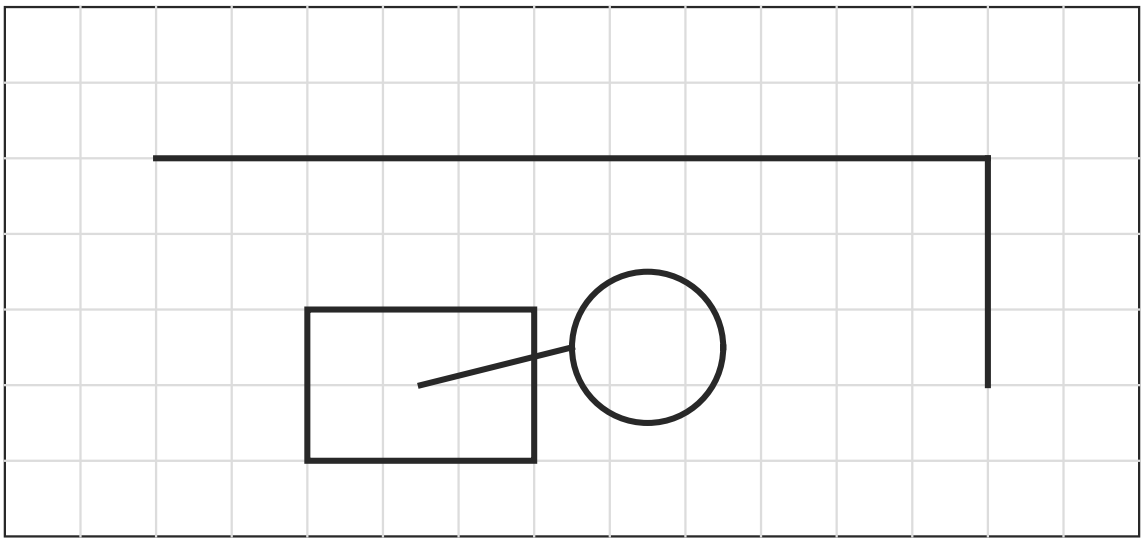
2.0 Core Architecture (Section 2.1)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.10: Kinematic Assembly Detail

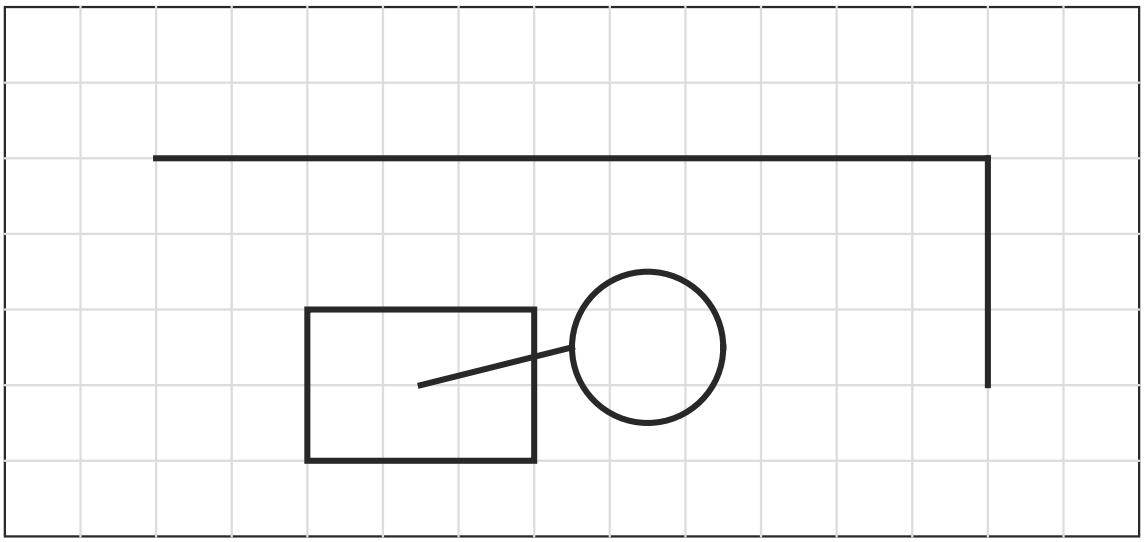


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.2)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.11: Kinematic Assembly Detail



Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.3)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

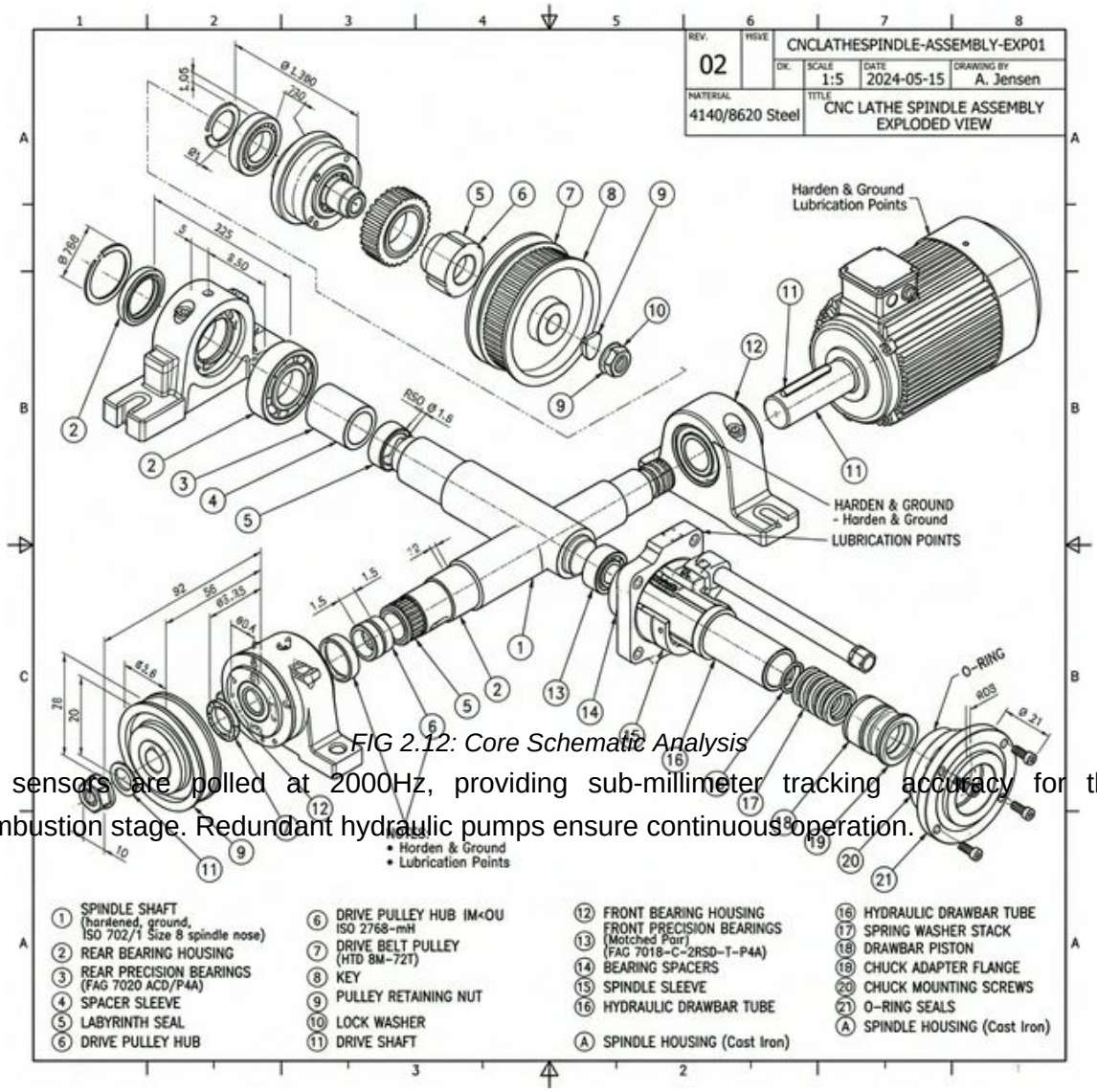


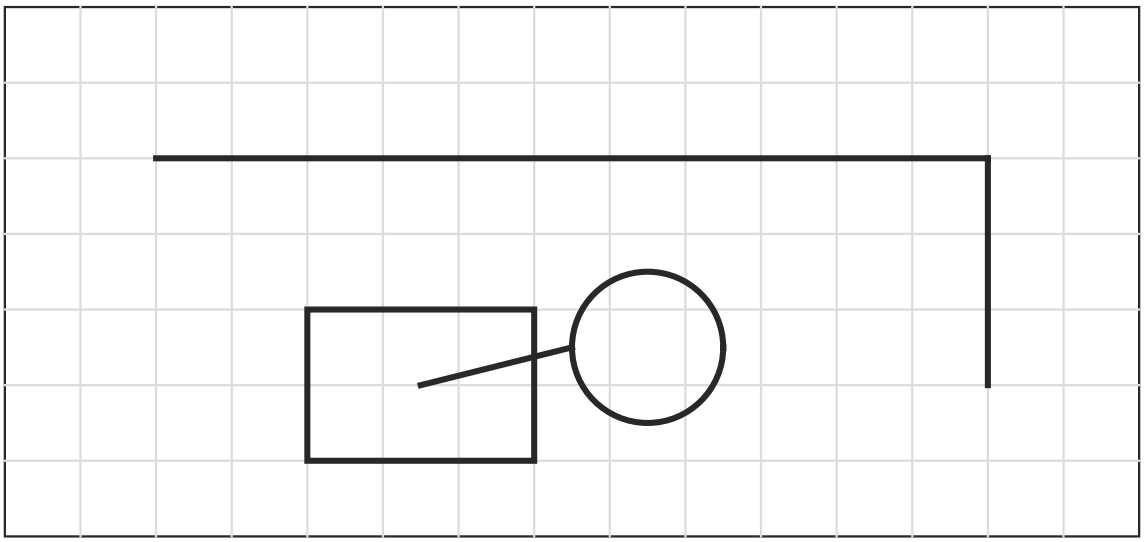
FIG 2.12: Core Schematic Analysis

Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.4)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.13: Kinematic Assembly Detail

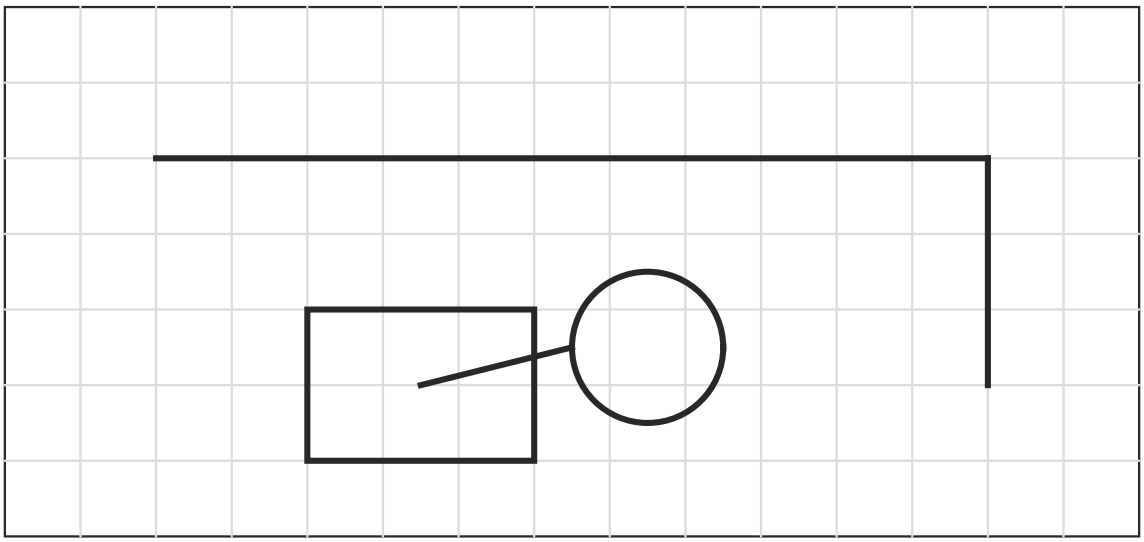


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.5)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.14: Kinematic Assembly Detail

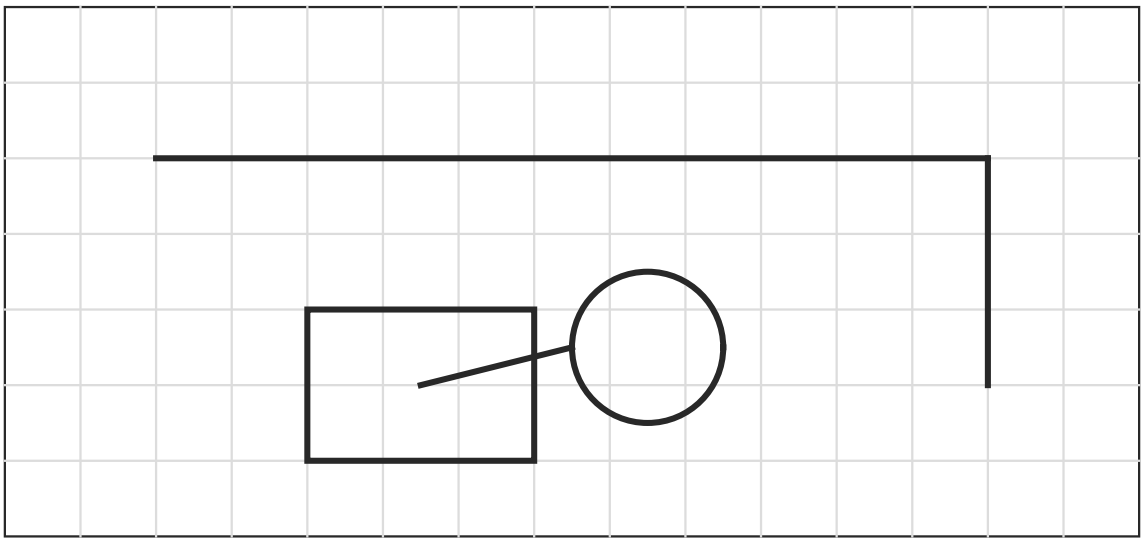


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.6)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.15: Kinematic Assembly Detail

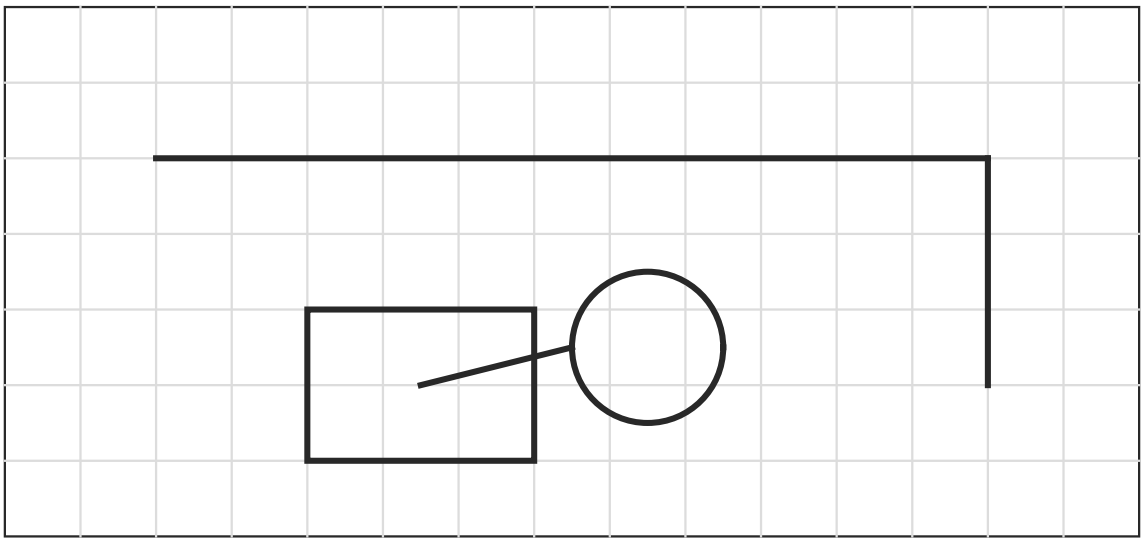


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.7)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.16: Kinematic Assembly Detail

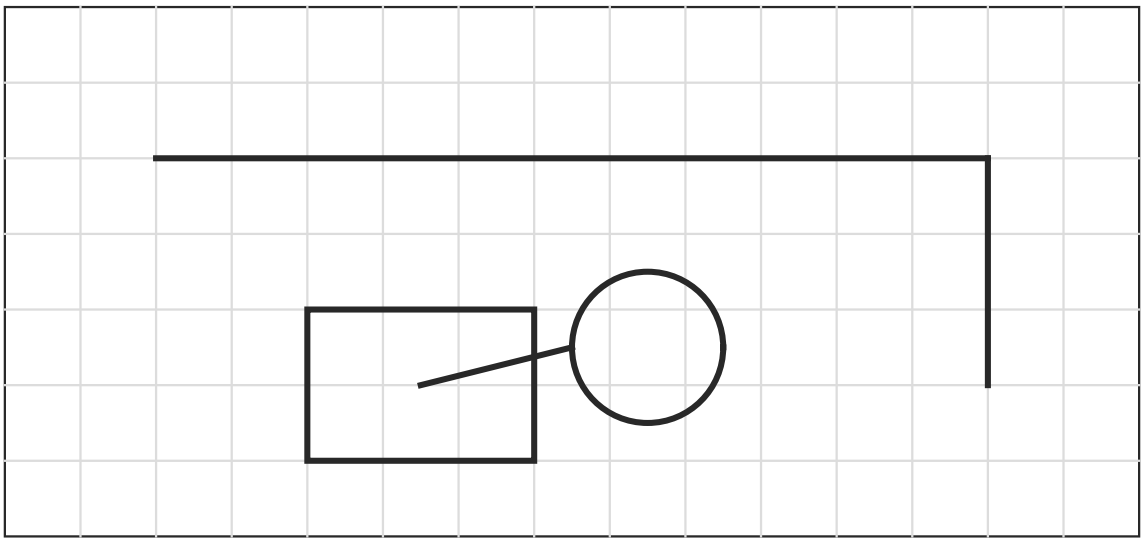


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.8)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.17: Kinematic Assembly Detail

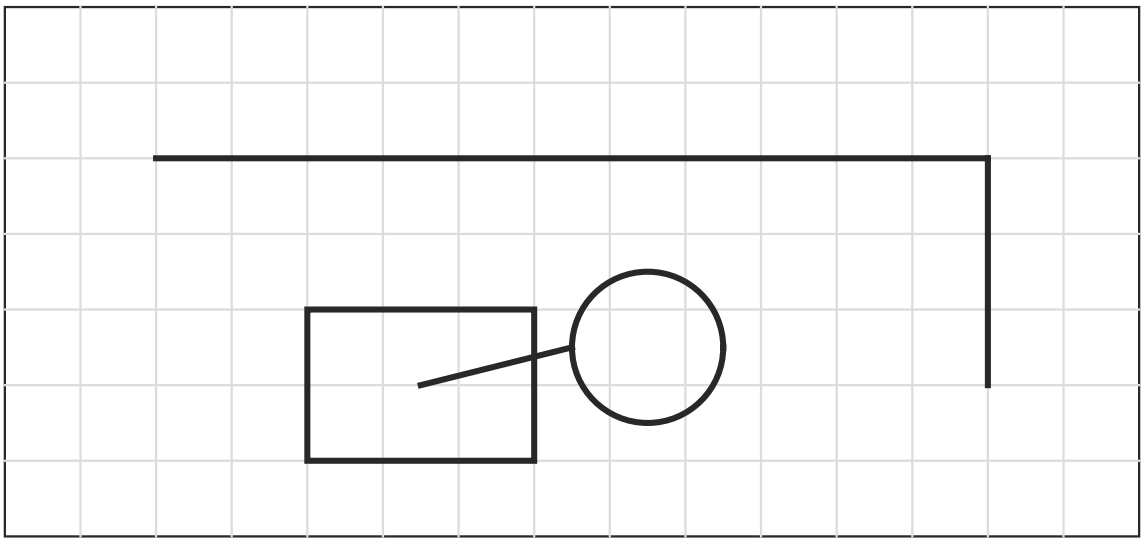


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.9)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.18: Kinematic Assembly Detail

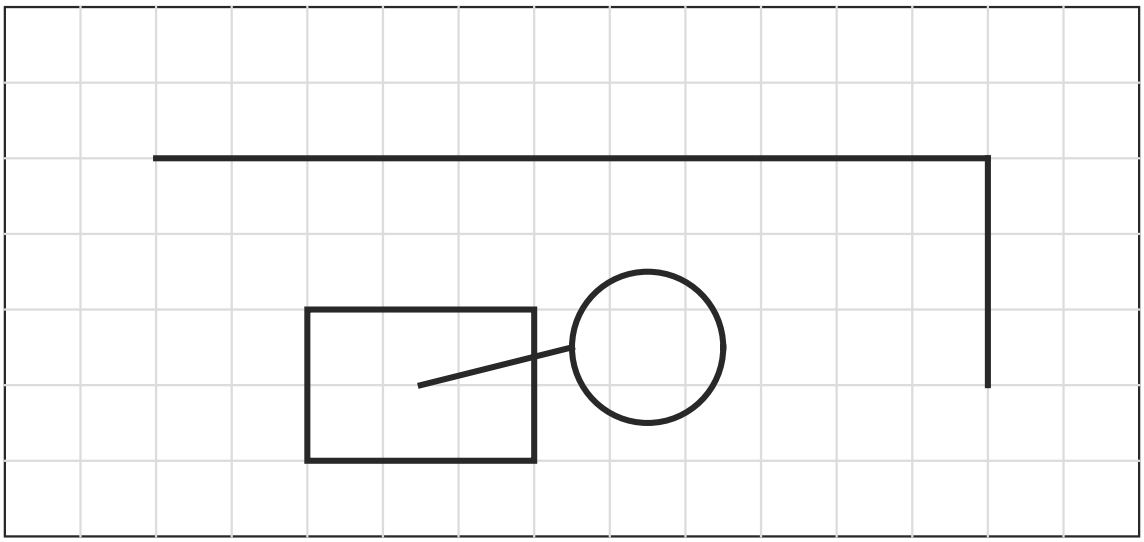


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.10)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.19: Kinematic Assembly Detail

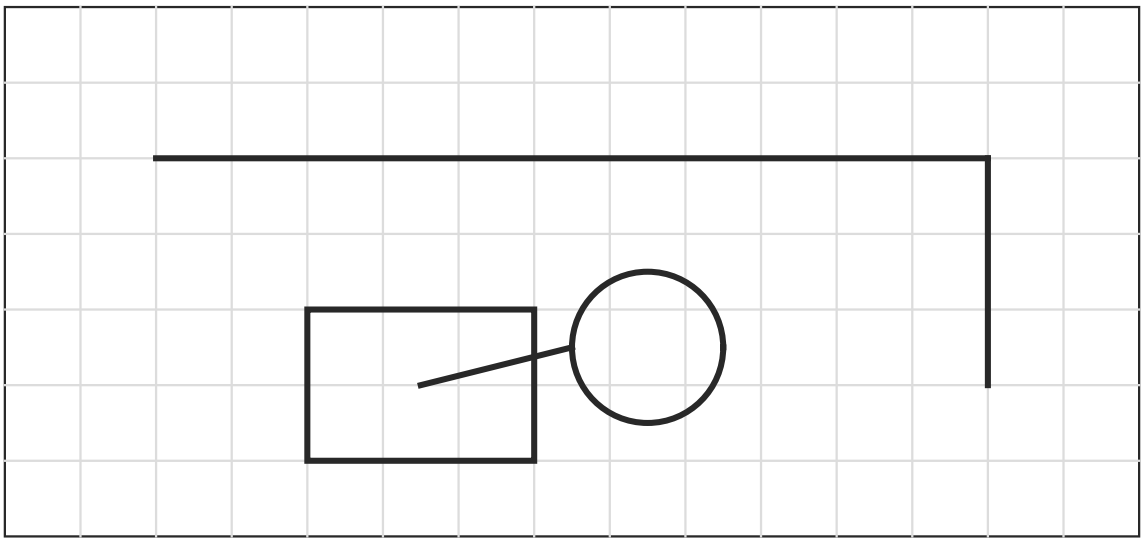


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.11)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.20: Kinematic Assembly Detail

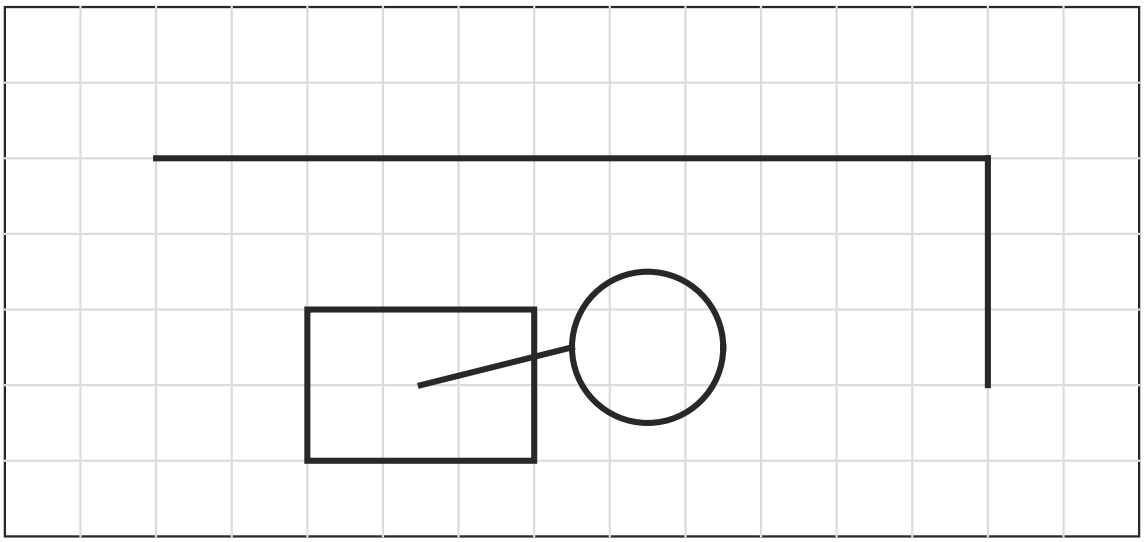


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.12)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.21: Kinematic Assembly Detail

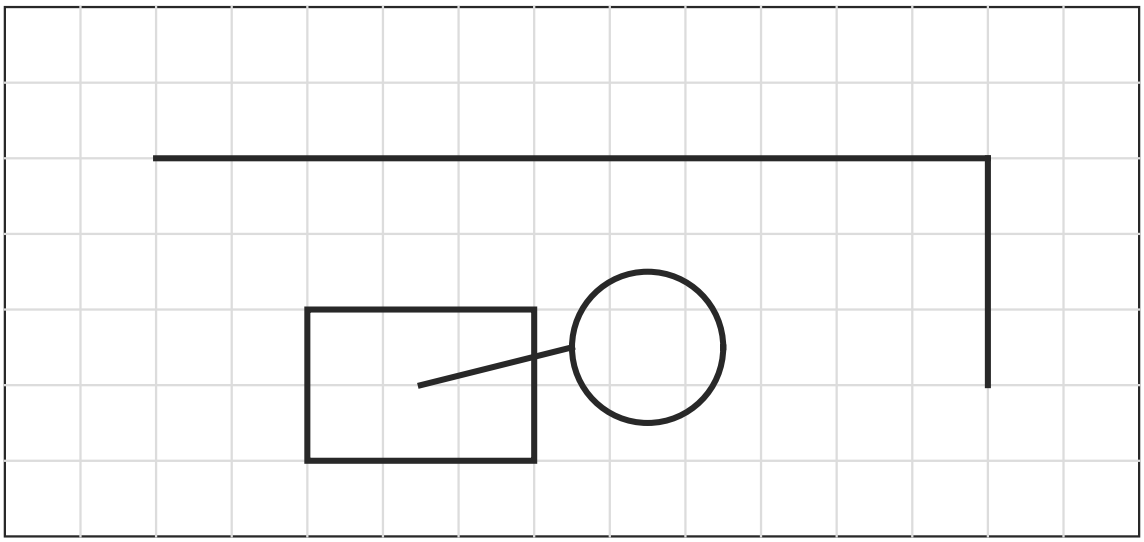


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.13)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.22: Kinematic Assembly Detail

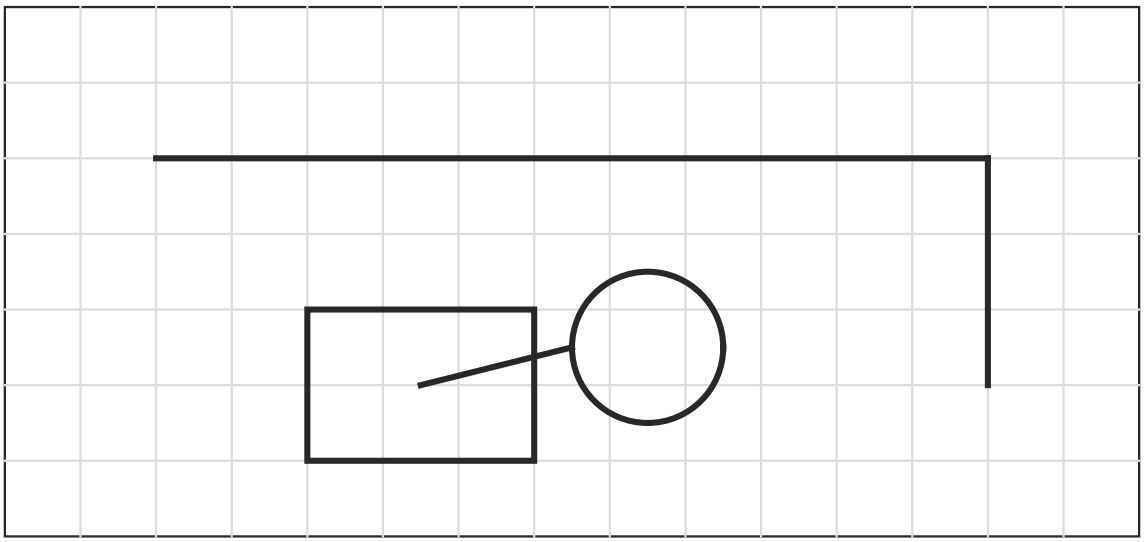


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.14)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.23: Kinematic Assembly Detail

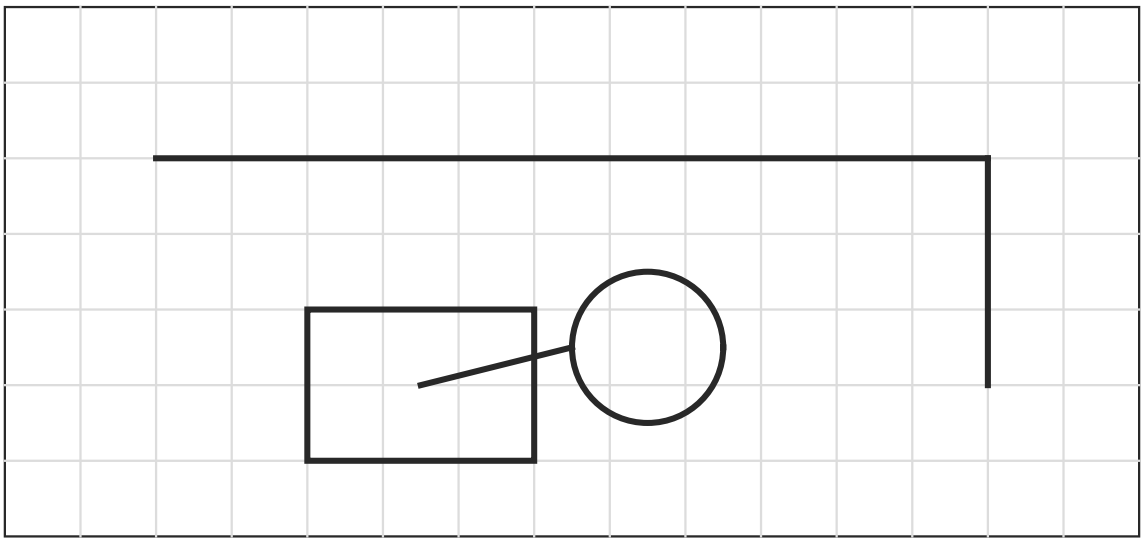


Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

2.0 Core Architecture (Section 2.15)

The kinematic drive incorporates dual planetary gears coupled to a hydrostatic bearing assembly. This ensures stiffness under heavy dynamic loads.

FIG 2.24: Kinematic Assembly Detail



Feedback sensors are polled at 2000Hz, providing sub-millimeter tracking accuracy for the central cutting/combustion stage. Redundant hydraulic pumps ensure continuous operation.

3.0 Installation Procedures (Step 3.1)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.2)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
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M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.3)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

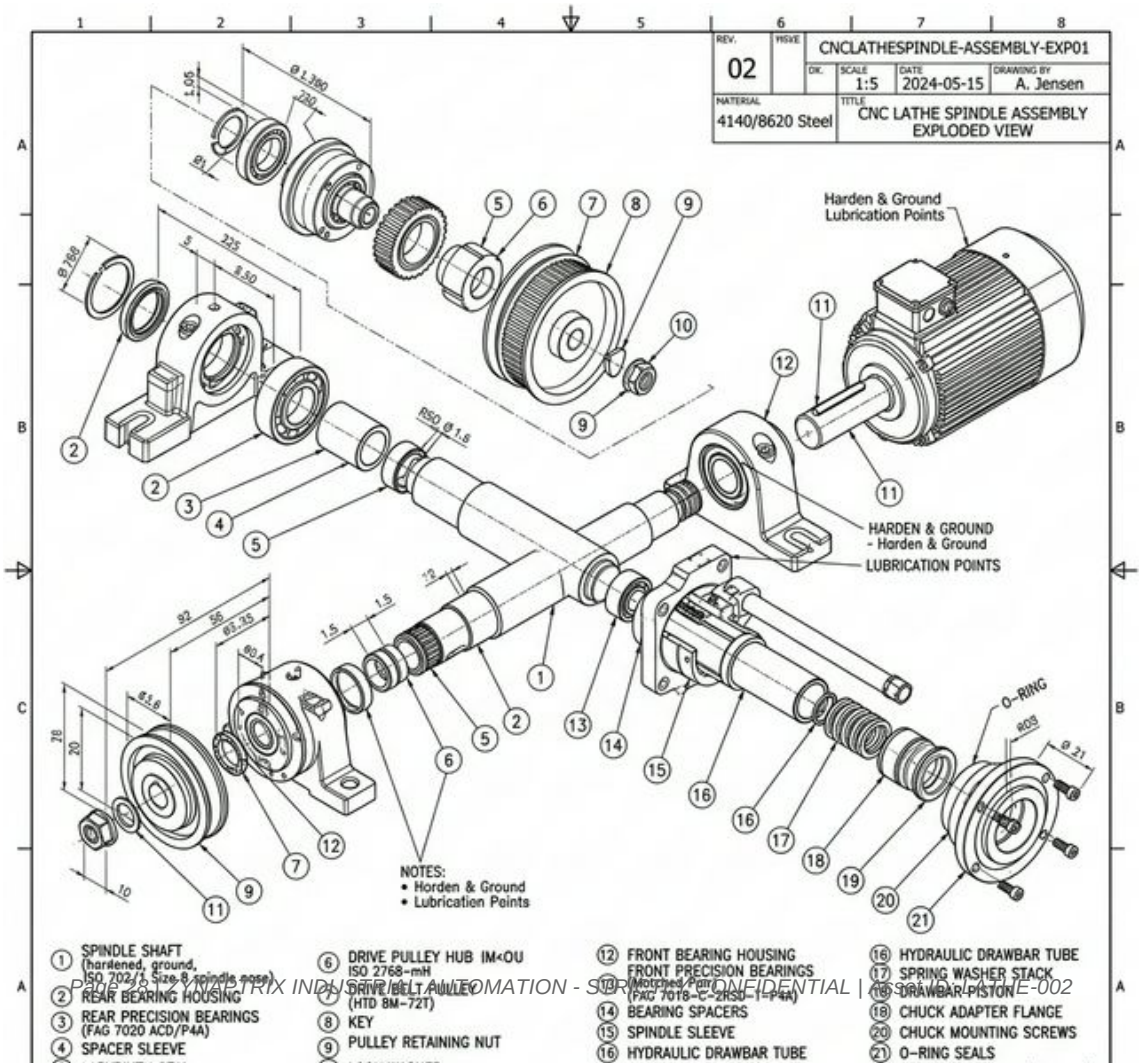
Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.4)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.



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3.0 Installation Procedures (Step 3.5)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.6)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.7)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.8)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.9)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.10)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.11)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.12)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.13)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.14)

Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

3.0 Installation Procedures (Step 3.15)

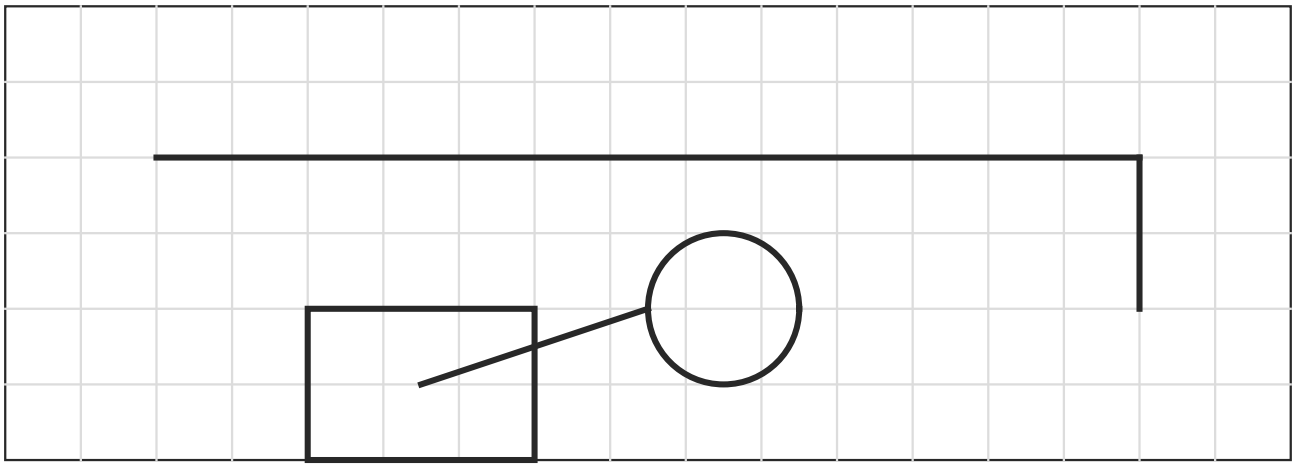
Torque Specifications

Bolt Type	Location	Torque (Nm)	Tool Req.
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench
M24x3.0	Base Anchor	450 Nm	Hydraulic wrench

Level the platform using a precision laser transit. Maximum allowed variance is 0.05mm per meter. Use Zynaptrix shim kits for fine adjustments.

4.0 AI Control System & HMI (Section 4.1)

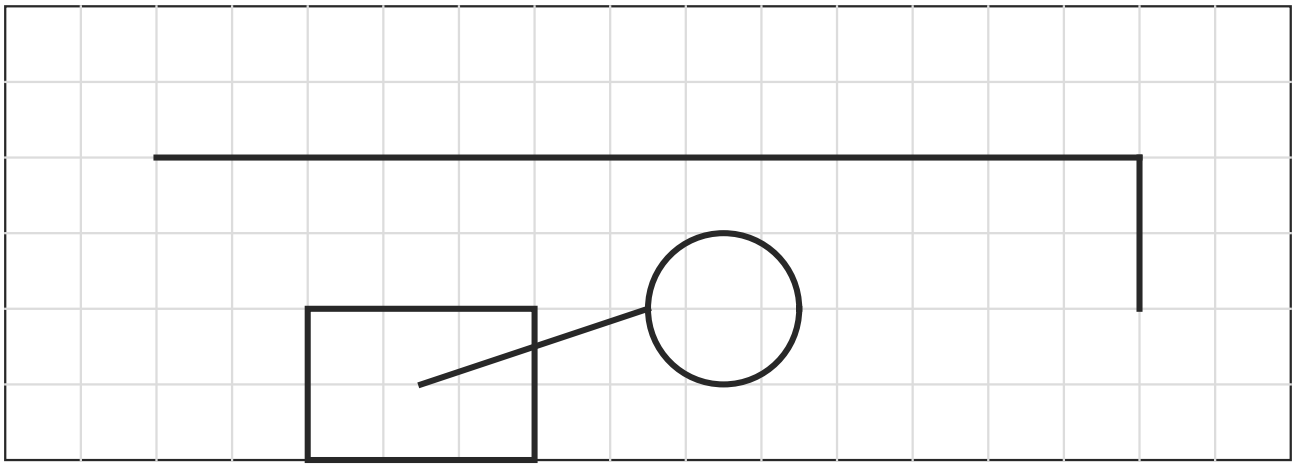
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

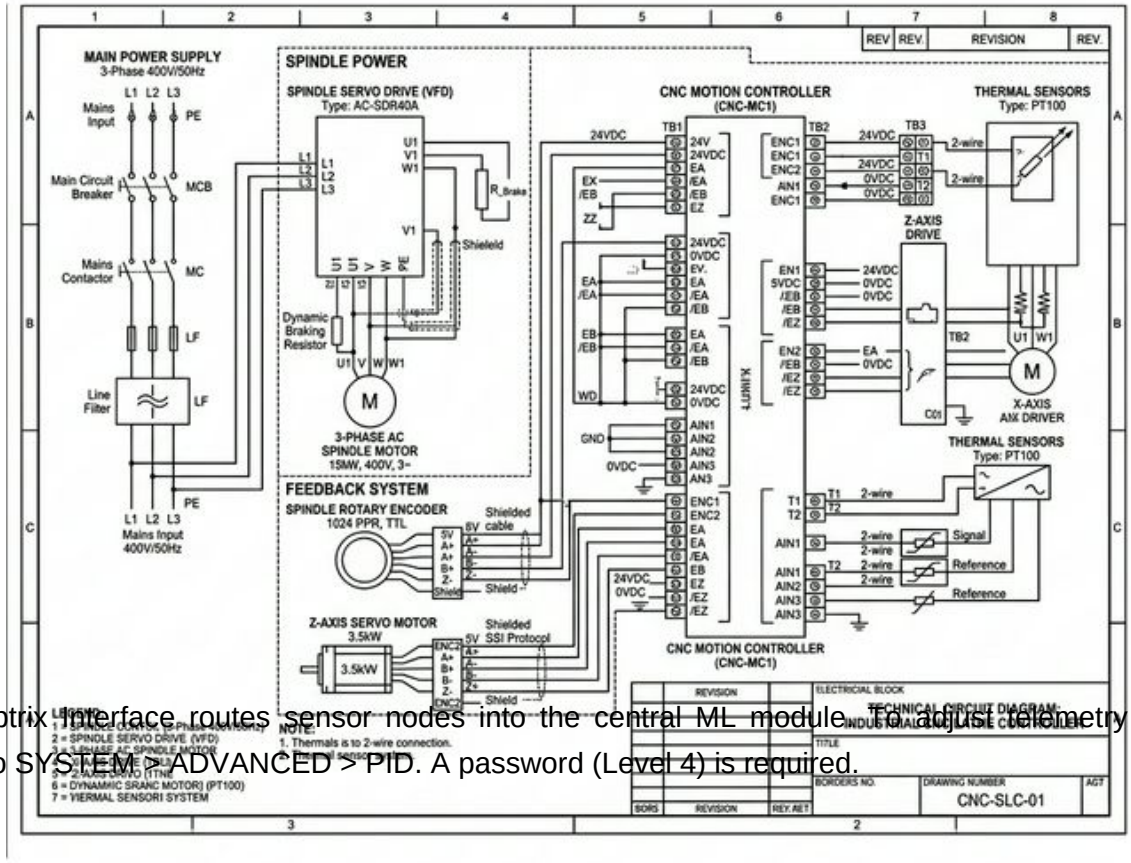
4.0 AI Control System & HMI (Section 4.2)

UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

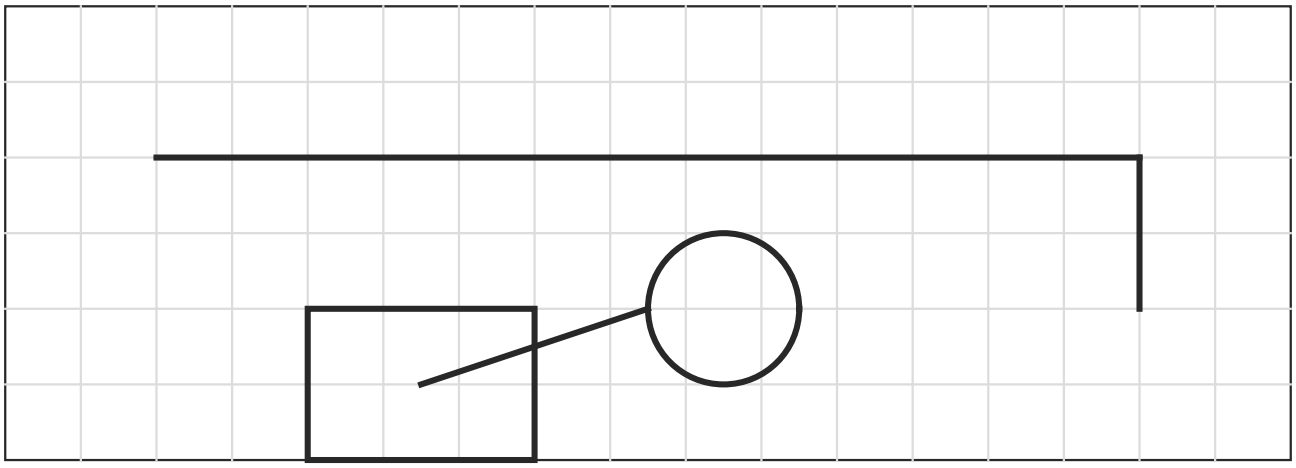
4.0 AI Control System & HMI (Section 4.3)



The Zynaptrix interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.4)

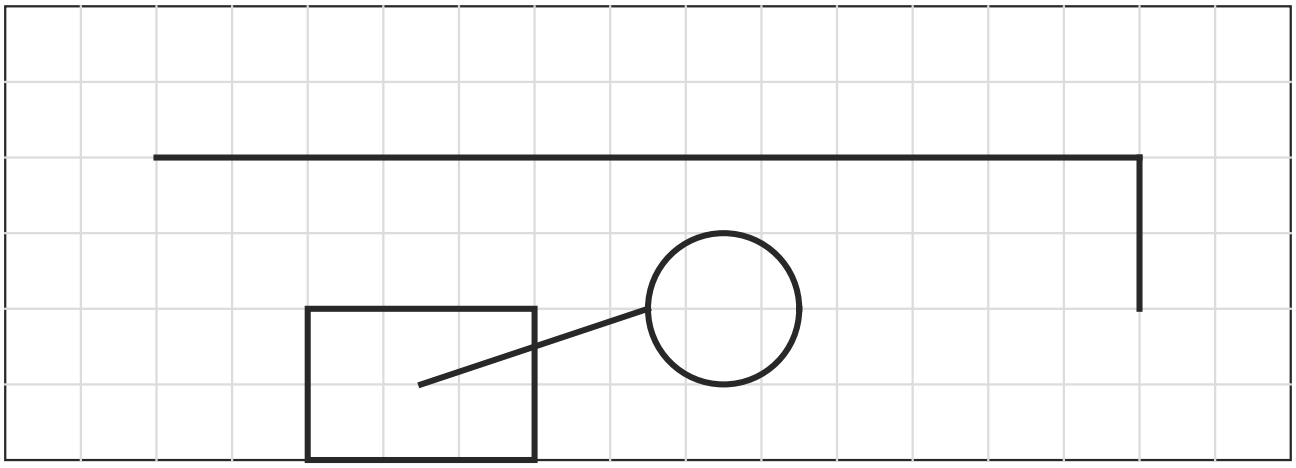
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.5)

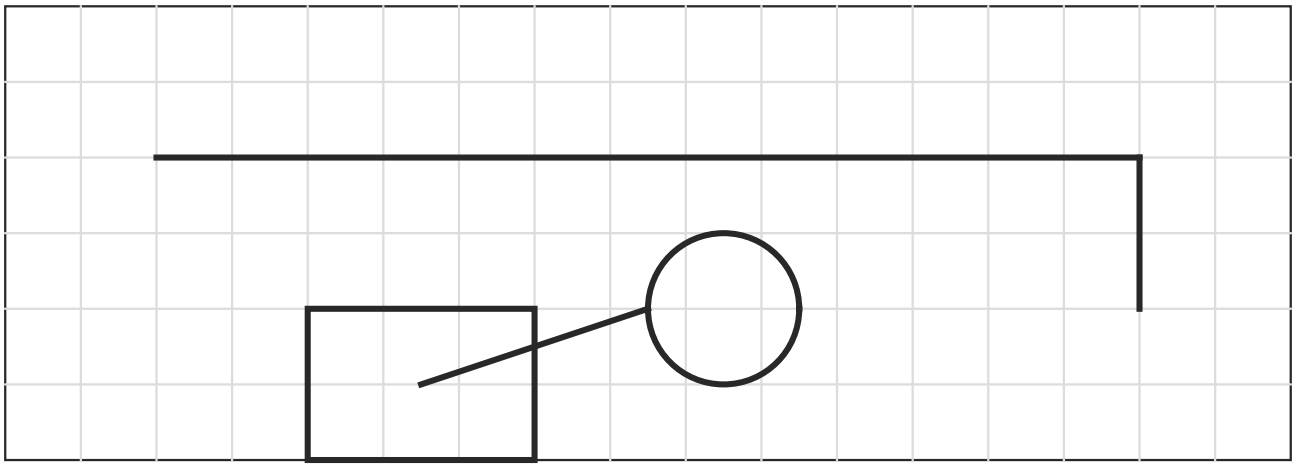
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.6)

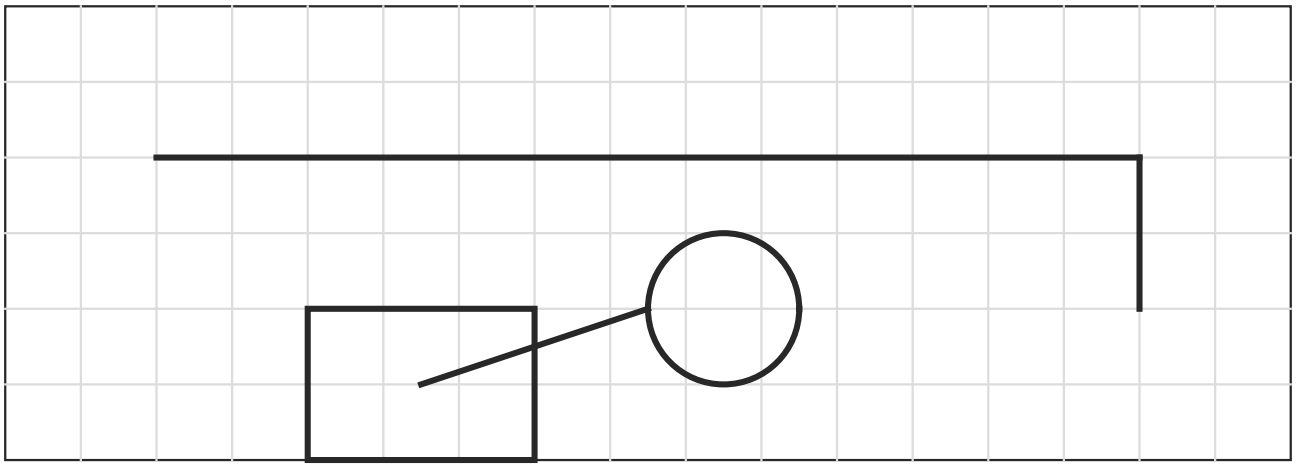
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.7)

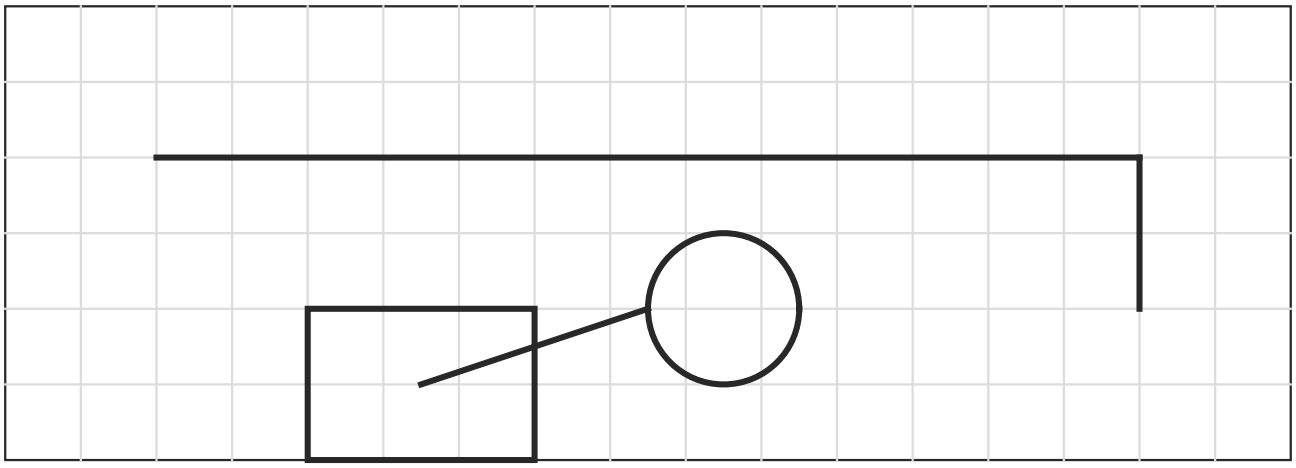
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.8)

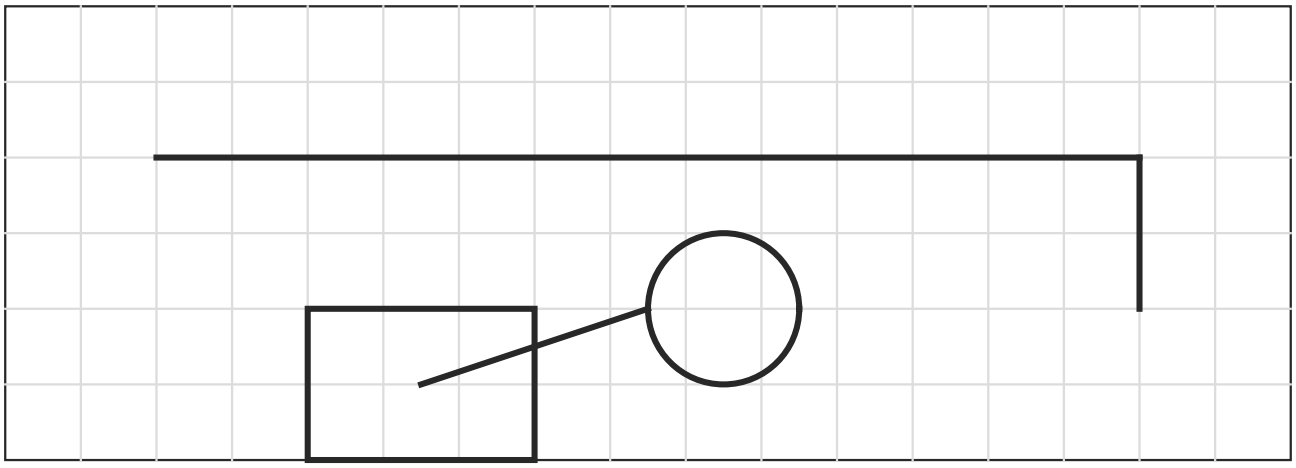
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.9)

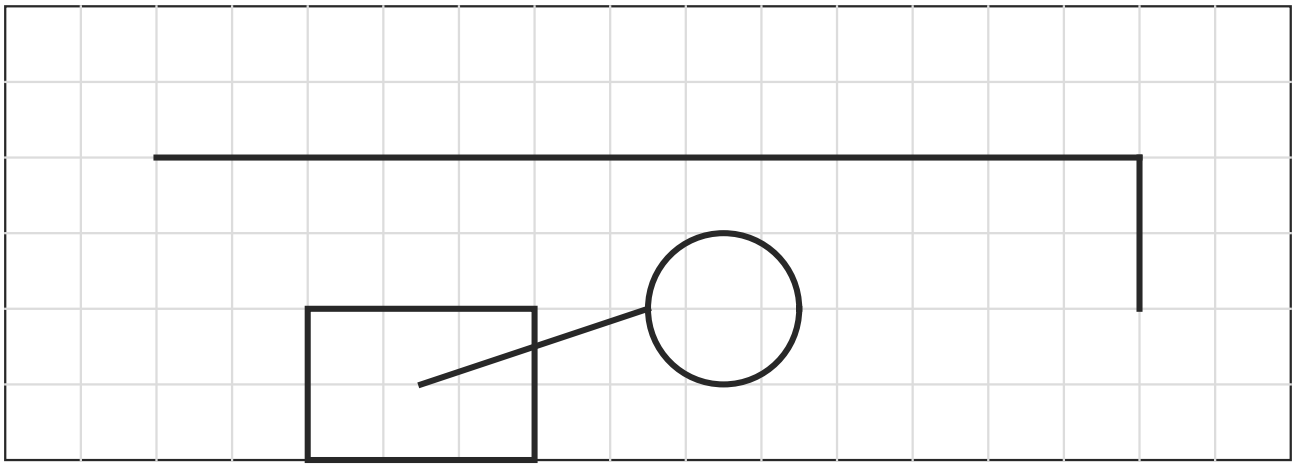
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.10)

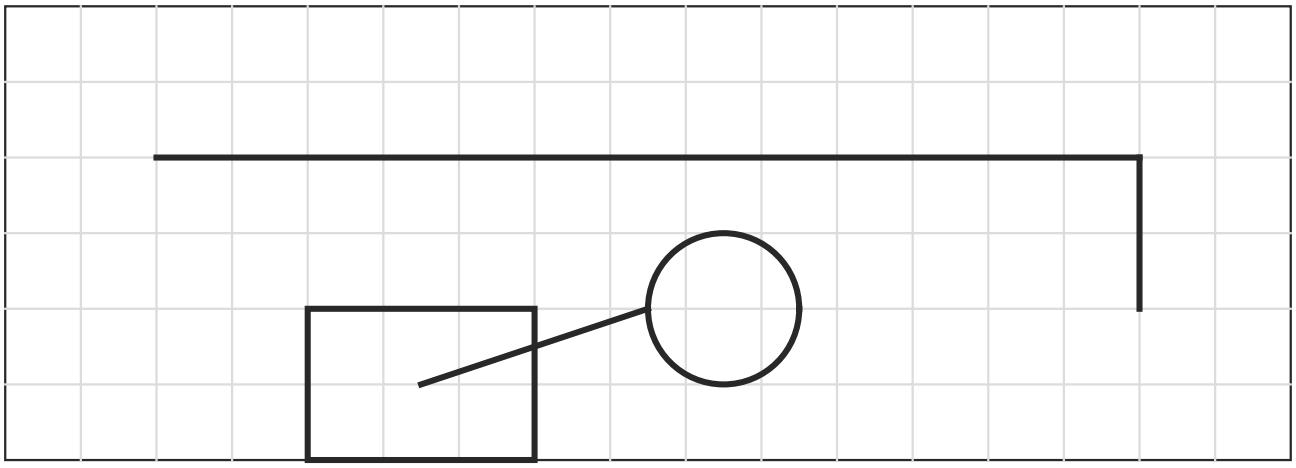
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.11)

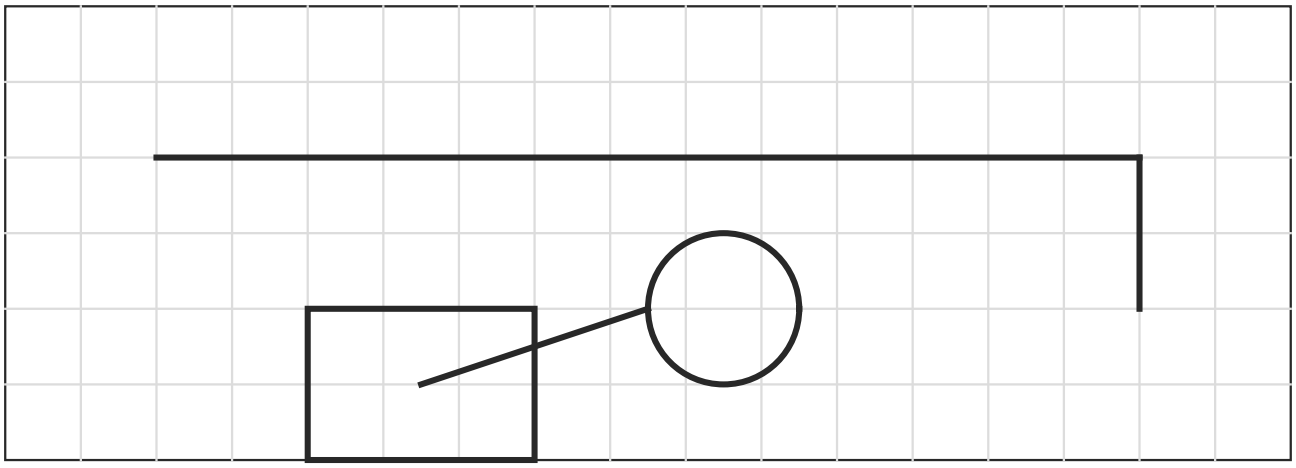
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.12)

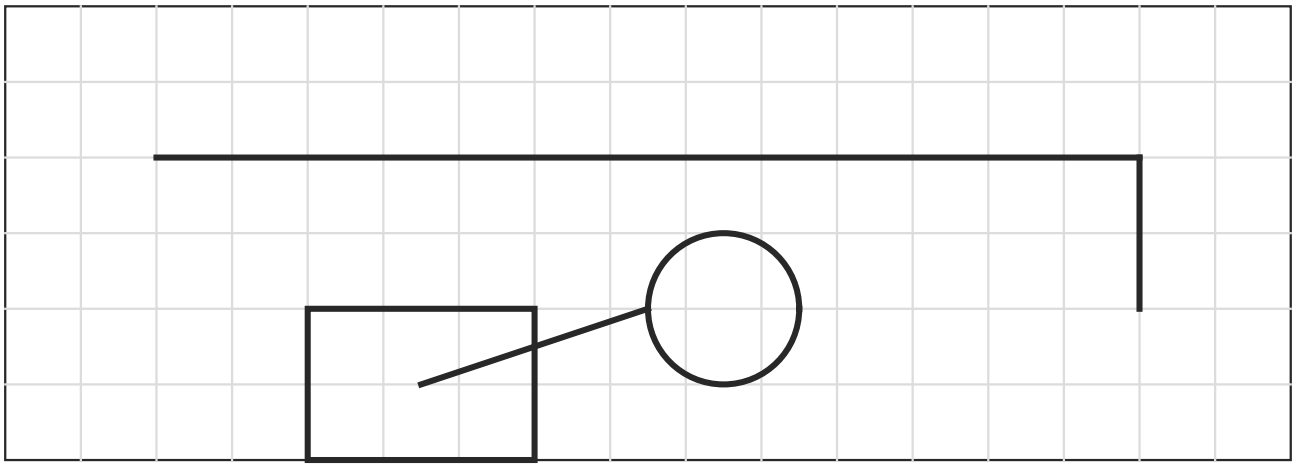
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.13)

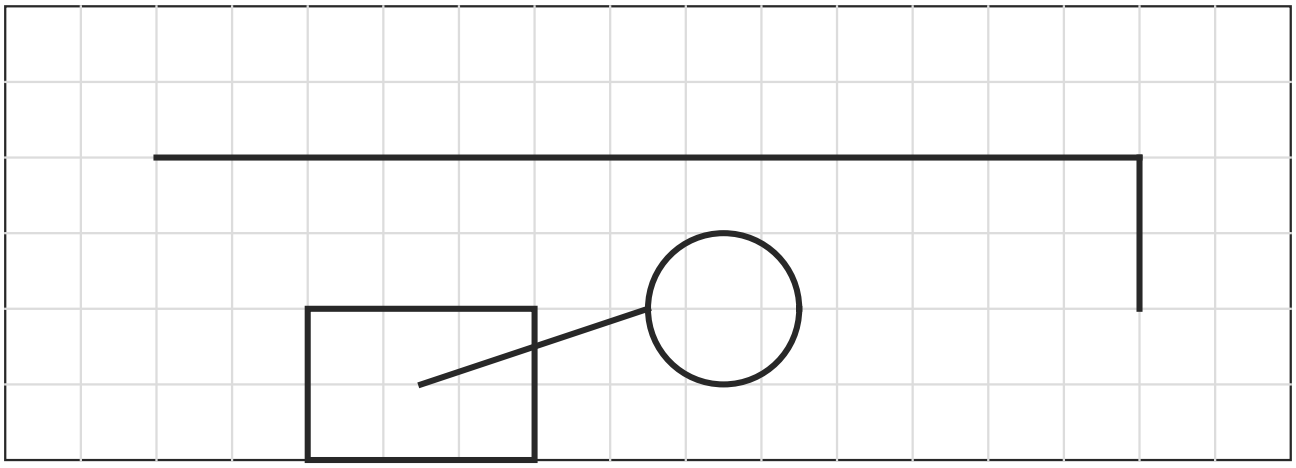
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.14)

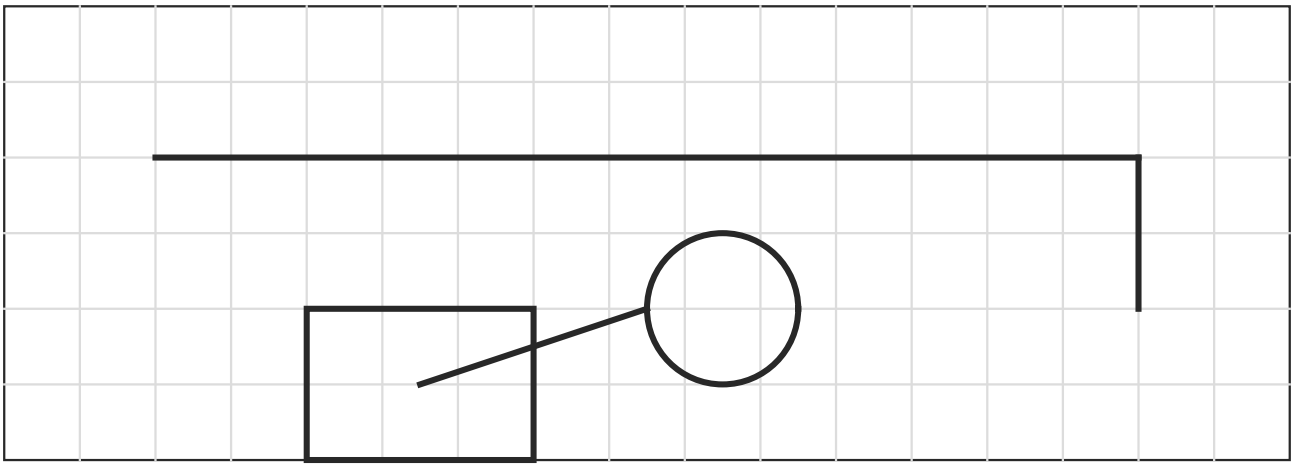
UI WIREFRAME: MAIN DASHBOARD



The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

4.0 AI Control System & HMI (Section 4.15)

UI WIREFRAME: MAIN DASHBOARD

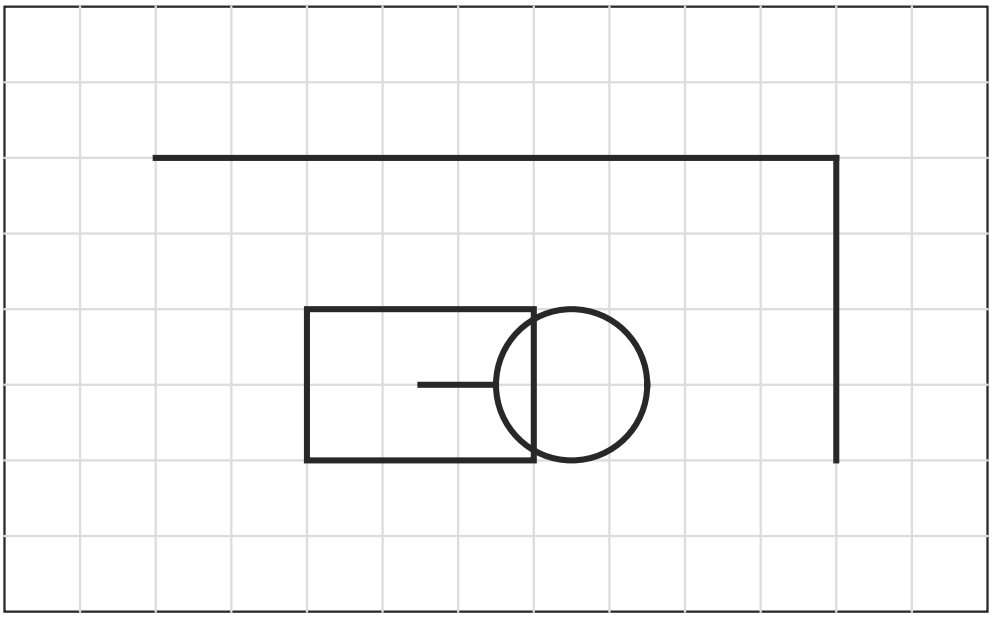


The Zynaptrix Interface routes sensor nodes into the central ML module. To adjust telemetry thresholds, navigate to SYSTEM > ADVANCED > PID. A password (Level 4) is required.

5.0 Advanced Diagnostics (Ref 5.1)

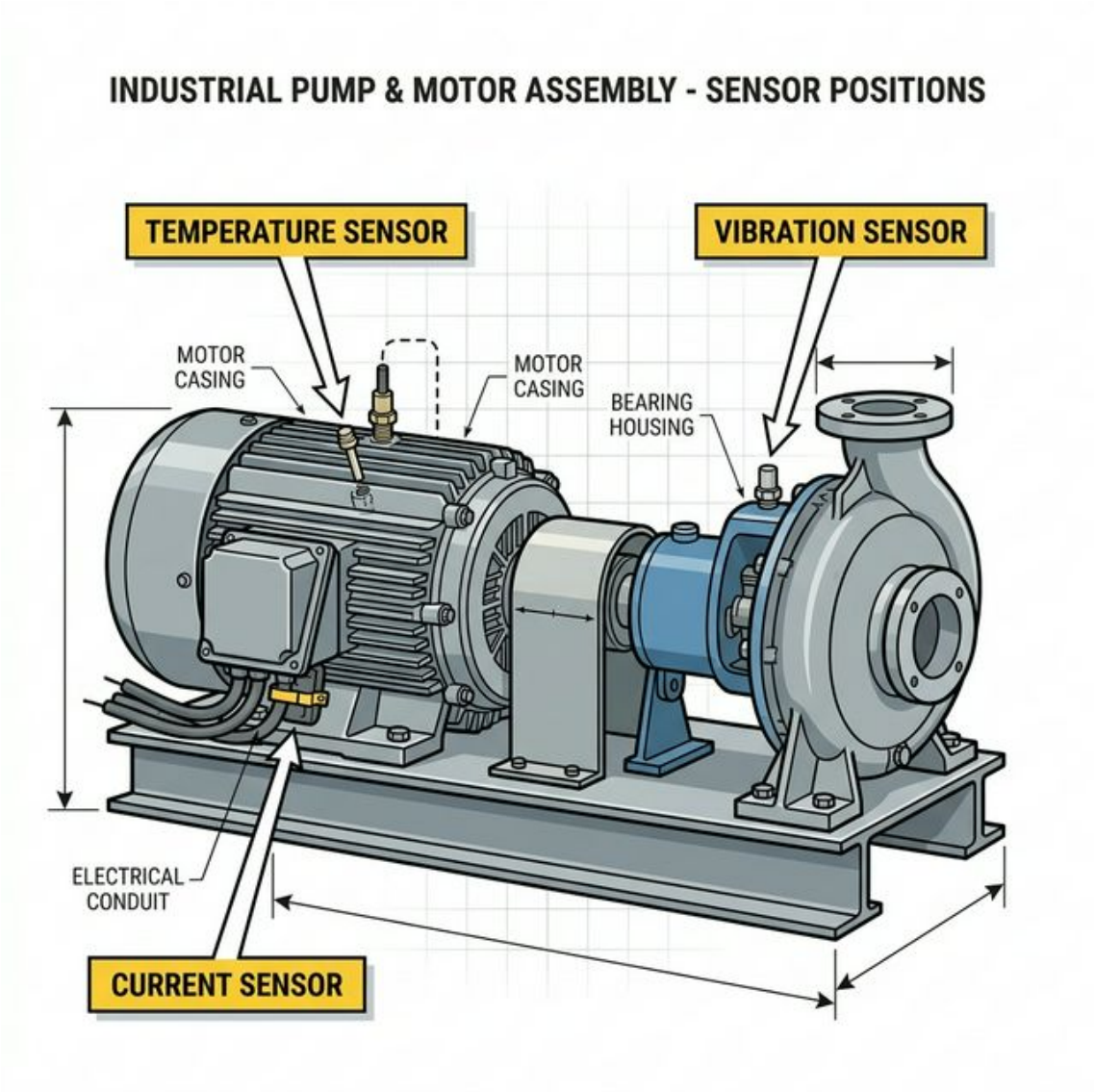
Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.2)

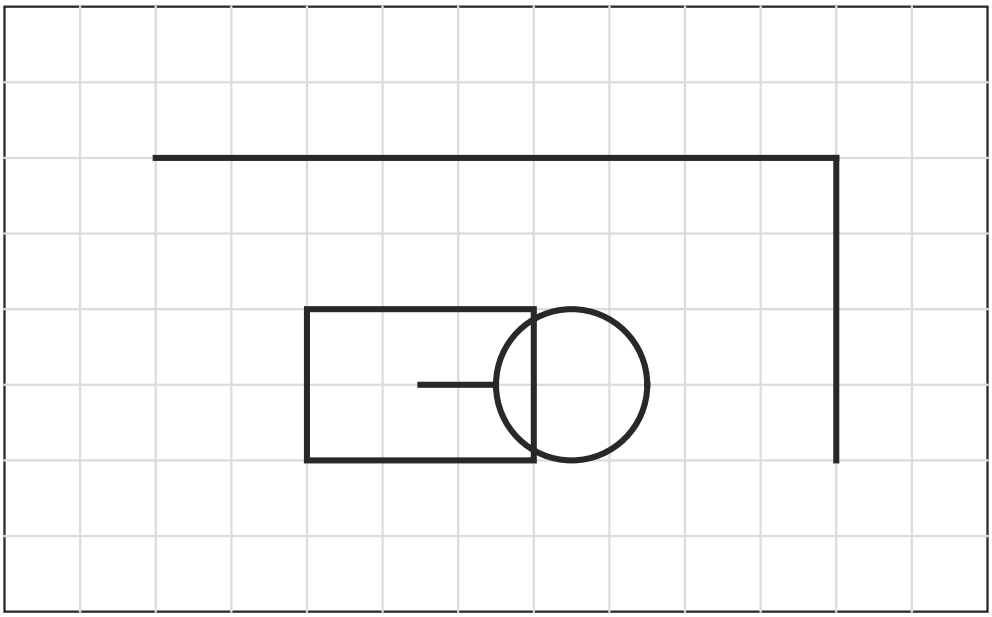
Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.



5.0 Advanced Diagnostics (Ref 5.3)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.4)

[!] DANGER

FAULT: LATHE-ERR-404 (Spindle Motor Stall)

Trigger: VFD current limit exceeded (>12A) or encoder desync.

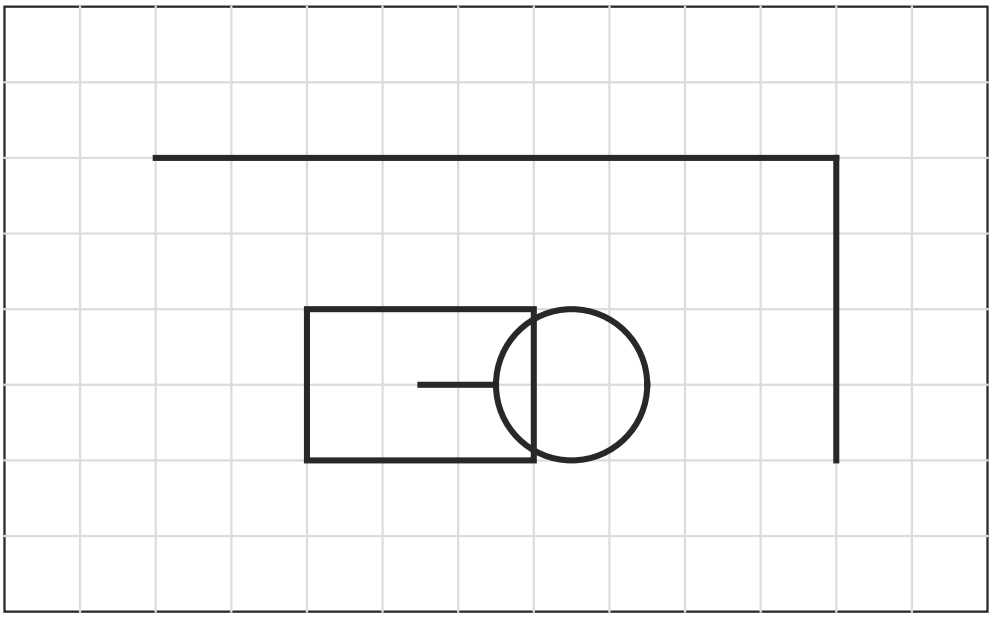
Procedure:

1. E-STOP immediately.
2. Verify cutting tool is not bound in workpiece.

5.0 Advanced Diagnostics (Ref 5.5)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

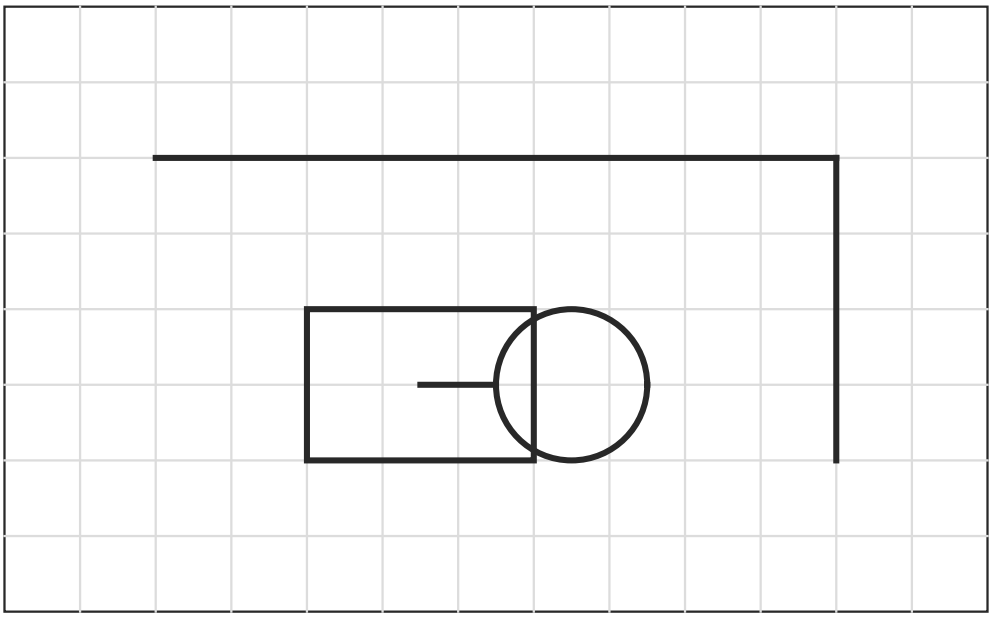
DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.6)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

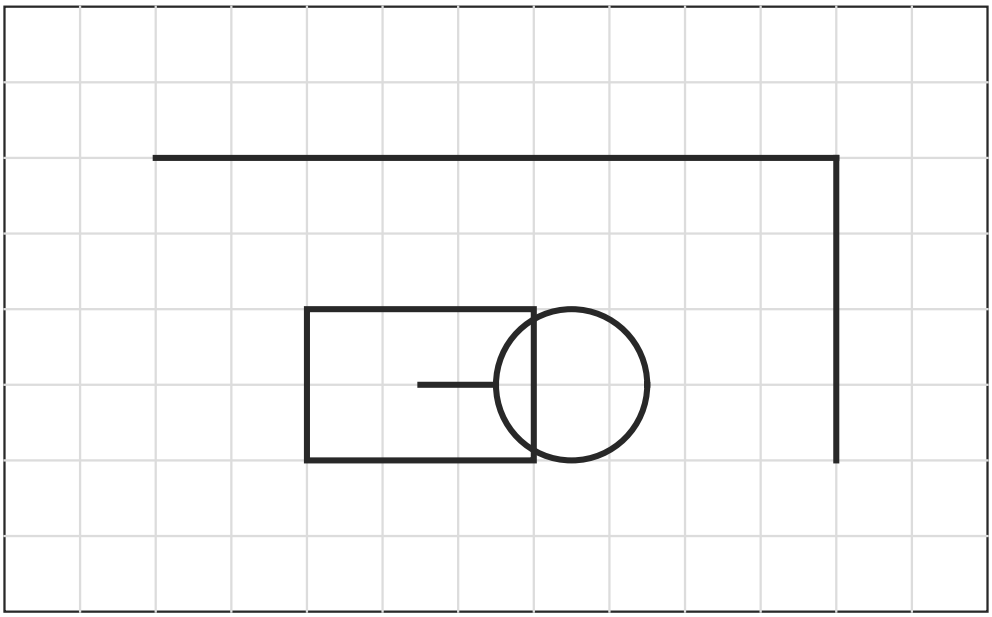
DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.7)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

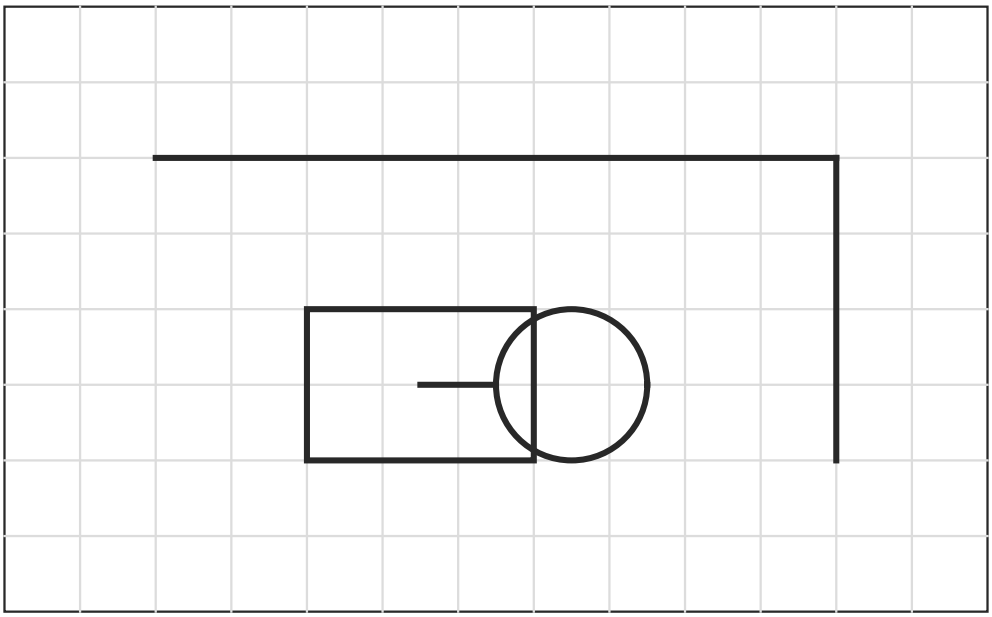
DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.8)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

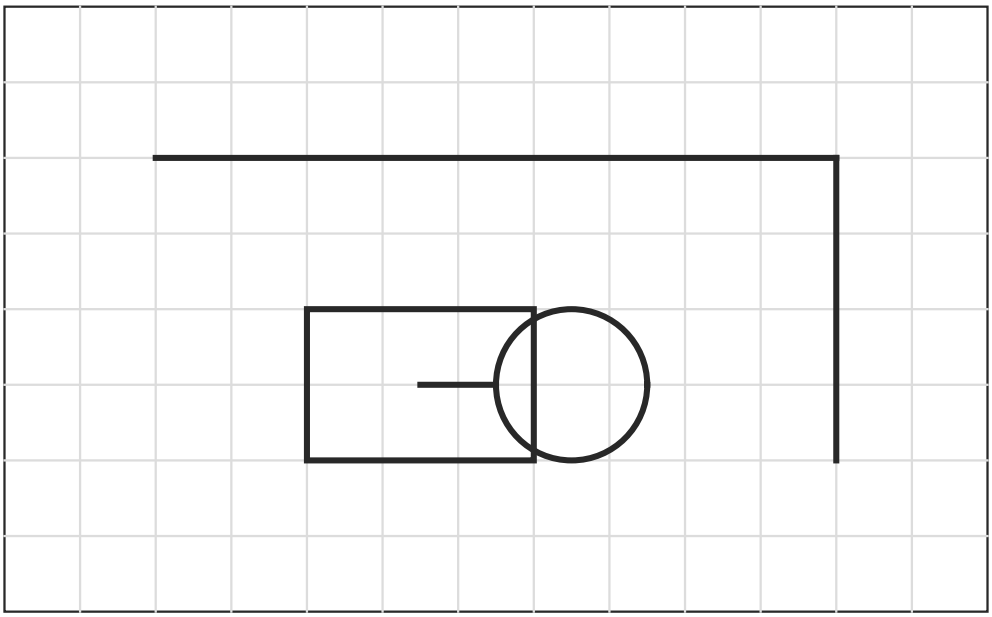
DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.9)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

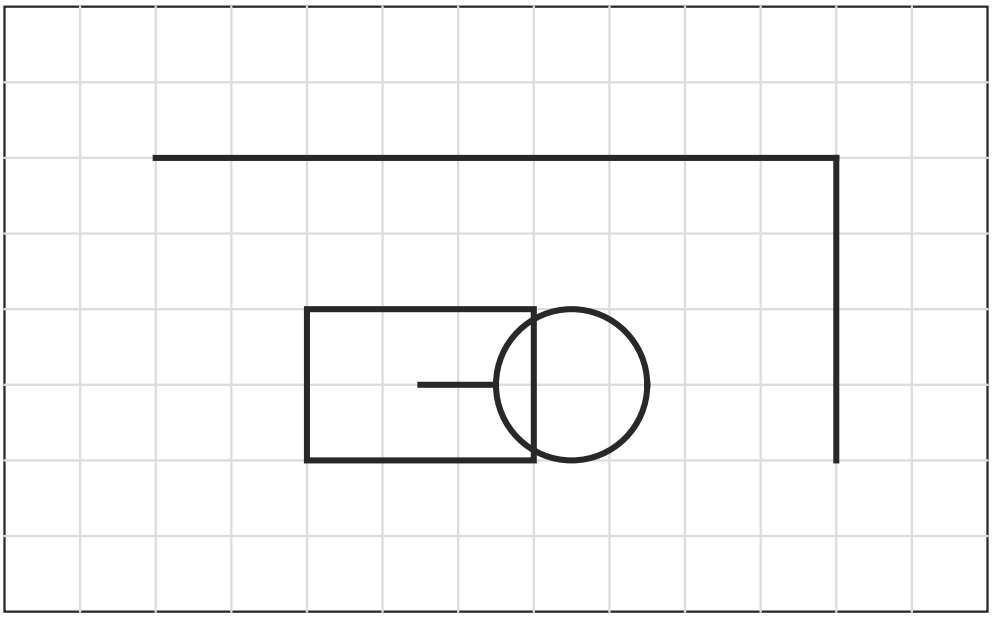
DIAGNOSTIC FLOWCHART: SENSOR FAILURE



5.0 Advanced Diagnostics (Ref 5.10)

Standard troubleshooting begins with analyzing the log files located in /var/log/zynaptrix/. Search for instances of 'warn' or 'crit' within the main process loop.

DIAGNOSTIC FLOWCHART: SENSOR FAILURE

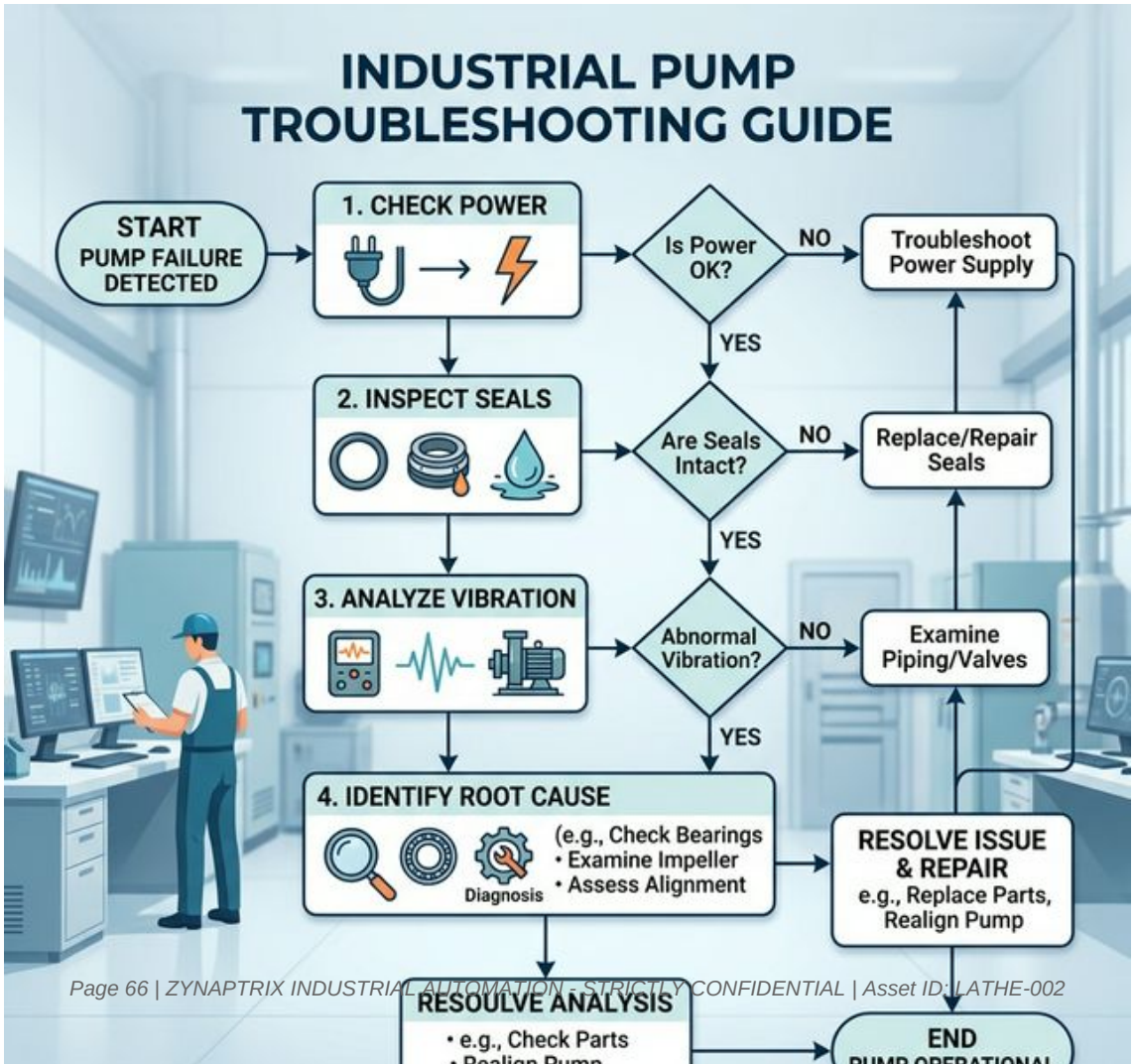


6.0 Maintenance Operations (Log 65)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 66)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X



6.0 Maintenance Operations (Log 67)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 68)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 69)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 70)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 71)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 72)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 73)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

6.0 Maintenance Operations (Log 74)

ID	Task	Interval	Consumable PN
PM-100	Inspect seals	500 hours	ZYN-SEAL-X
PM-200	Inspect seals	500 hours	ZYN-SEAL-X
PM-300	Inspect seals	500 hours	ZYN-SEAL-X
PM-400	Inspect seals	500 hours	ZYN-SEAL-X
PM-500	Inspect seals	500 hours	ZYN-SEAL-X
PM-600	Inspect seals	500 hours	ZYN-SEAL-X
PM-700	Inspect seals	500 hours	ZYN-SEAL-X
PM-800	Inspect seals	500 hours	ZYN-SEAL-X
PM-900	Inspect seals	500 hours	ZYN-SEAL-X
PM-1000	Inspect seals	500 hours	ZYN-SEAL-X
PM-1100	Inspect seals	500 hours	ZYN-SEAL-X
PM-1200	Inspect seals	500 hours	ZYN-SEAL-X
PM-1300	Inspect seals	500 hours	ZYN-SEAL-X
PM-1400	Inspect seals	500 hours	ZYN-SEAL-X

7.0 Exhaustive Parts Catalog (Sheet 1)

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LATHE-002-0754 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0759 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0762 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0763 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0771 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0772 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0773 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0774 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0781 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0782 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0783 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0784 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0788 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0789 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 2)

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LATHE-002-0760 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
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LATHE-002-0762 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0763 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0773 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0783 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0786 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
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LATHE-002-0788 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0789 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0792 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0793 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0795 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0796 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0797 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0798 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0799 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 3)

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LATHE-002-0770 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
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LATHE-002-0772 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0773 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0774 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0775 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0776 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0777 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0778 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0782 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0783 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0784 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0785 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0786 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0787 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
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LATHE-002-0803 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0804 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0808 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0809 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 4)

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LATHE-002-0783 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0784 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0786 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0787 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0788 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0789 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0790 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0791 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0792 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0793 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0814 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0818 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0819 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 5)

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LATHE-002-0793 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0794 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0795 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0796 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 6)

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LATHE-002-0803 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0806 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0807 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0808 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0809 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0813 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0814 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0815 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
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LATHE-002-0818 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0819 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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LATHE-002-0823 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
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LATHE-002-0838 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0839 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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7.0 Exhaustive Parts Catalog (Sheet 7)

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LATHE-002-0813 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0814 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0815 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0816 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0817 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0818 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0819 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0820 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0821 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0822 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0823 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0824 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0825 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0826 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0827 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0828 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0829 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0830 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0831 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0832 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0833 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0834 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0835 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0836 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0837 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0838 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0839 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0840 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0841 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0842 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0843 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0844 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
LATHE-002-0845 | SUB-ASSEMBLY BLOCK | Qty: 1 | REV C
LATHE-002-0846 | SUB-ASSEMBLY BLOCK | Qty: 2 | REV C
LATHE-002-0847 | SUB-ASSEMBLY BLOCK | Qty: 3 | REV C
LATHE-002-0848 | SUB-ASSEMBLY BLOCK | Qty: 4 | REV C
LATHE-002-0849 | SUB-ASSEMBLY BLOCK | Qty: 5 | REV C
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